Compactness :→

Defn: (X,d) metric space, $E \subseteq X$ we say that E is compact in X iff for every open cover $\mathcal{F}$ of E there is a finite $\mathcal{E} \subseteq \mathcal{F}$ s.t. $\cup \mathcal{E} \supseteq E$.

Def: Suppose (X,d) is a metric space and $E \subseteq X$ An open cover of E is a ~~finitely~~ family of $\mathcal{F}$ of open subsets of X, s.t. $E \subseteq \cup \mathcal{F}$.

Suppose $\mathcal{F}$ is an open cover of E and $\mathcal{E} \subseteq \mathcal{F}$ in a subcover of E if $\cup \mathcal{E} \supseteq E$.

Ex: ① (X,d) metric space & $E \subseteq X$

Proof If for every open cover $\mathcal{F}$ of E s.t. each member of $\mathcal{F}$ is an open ball in X there is a finite $\mathcal{E} \subseteq \mathcal{F}$ s.t. $\cup \mathcal{E} \supseteq E$ then E is compact.

② $E \subseteq X$ is finite $\Rightarrow$ E is compact in X.

Proof Say $E = \{y_1, y_2, \ldots y_n\}$

let $\mathcal{F}$ be an open cover of E

Then ⓐ Each $U \in \mathcal{F}$ is an open set in X.

ⓑ $\cup \mathcal{F} \supseteq E$

Choose $U_1 \in \mathcal{F}$ s.t. $y_1 \in U_1$.

$U_2 \in \mathcal{F}$ $y_2 \in U_2$

choose $U_k \in \mathcal{F}$ s.t. $y_n \in U_n$

$\mathcal{E} = \{U_1, \ldots U_n\} \subseteq \mathcal{F}$

$\bigcup \mathcal{E} = \bigcup_{k=1}^{n} U_k \supseteq E$.

Theorem: → $[a, b]$ is compact in $\mathbb{R}$

$(\forall\ a < b)$

Proof: Let $\mathcal{F}$ be an open cover of $[a, b]$

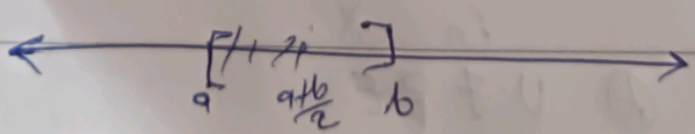
Key observation:- If A is a cover of EUF & no finite BCA is a cover of EUF then either no finite BCA is a cover of E.

OR No finite BCA is a cover of F

↓ Assume both are false → Prove by contradiction

↳ Towards a contradiction, suppose there is no finite $\mathcal{E} \subseteq \mathcal{F}$ s.t.

$\bigcup \mathcal{E} \supseteq [a, b]$



$[a,b] = \left[a, \frac{a+b}{2}\right] \cup \left[\frac{a+b}{2}, b\right]$

Either no finite $\mathcal{E} \subseteq \mathcal{F}$ s.t.

$\cup \mathcal{E} \supseteq \left[a, \frac{a+b}{2}\right]$

~~Either an finite $\mathcal{E} \subseteq \mathcal{F}$ s.t.~~

~~$\cup \mathcal{E} \supseteq \left[a, \frac{a+b}{2}\right]$~~

(or) ..... $\cup \mathcal{E} \supseteq \left[\frac{a+b}{2}, b\right]$

Inductively choose $\langle [a_n, b_n] : n \geq 1 \rangle$ s.t.

s.t.

(1) $[a_1, b_1] = [a, b]$

(2) $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$ &

$b_{n+1} - a_{n+1} = \frac{1}{2}(b_n - a_n)$

(3) $(\forall n \geq 1)$ (there is no finite $\mathcal{E} \subseteq \mathcal{F}$ s.t. $\cup \mathcal{E} \supseteq [a_n, b_n]$

Put $x = \sup(\{a_n : n \geq 1\})$

$= \inf(\{b_n : n \geq 1\})$

$\therefore [a,b] \times [c,d]$ is compact in $\mathbb{R}^2$

□

then $\bigwedge_{n \geq 1} [a_n, b_n] = \{x\}$

As $x \in [a,b] \subseteq \cup \mathcal{F}$, we can choose $V \in \mathcal{F}$ s.t. $x \in V$

As V is open in $\mathbb{R}$, there exists $\eta > 0$ s.t.

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Thm:→ Let (X,d) be a metric space $E \subseteq X$. then E is compared in X iff E is compact in E.

Proof: Suppose E is compact in X.

(a) Let $\mathcal{F}$ be a cover of E where each $V \in \mathcal{F}$ is open in E. For each $V \in \mathcal{F}$, fix $U_V \underset{\text{open}}{\subseteq} X$ s.t. $V = U_V \cap E$

Put $\mathcal{F}' = \{U_V : V \in \mathcal{F}\}$ Then $\mathcal{F}'$ is a cover of E via open sets in X. As E is compact in X.

$\exists\ \mathcal{E}' \subseteq \mathcal{F}'$ s.t. $\cup \mathcal{E}' \supseteq E$

$$\mathcal{E}' = \{U_{V_1}, U_{V_2} \ldots U_{V_n}\}$$

Then $\mathcal{E} = \{V_1 \ldots V_n\}$ is cover of E.

($\mathcal{E} \subseteq \mathcal{F}$ and, finite)

$V_1 = U_{V_1} \cap E$
$V_2 = U_{V_2} \cap E$
$\vdots$
$V_n = U_{V_n} \cap E$

(b) Next suppose E is compact in E.
Let $\mathcal{F}$ be a cover of E s.t. each $U \subset \mathcal{F}$ is open in X.

Consider $\mathcal{F}' = \{U \cap E : U \in \mathcal{F}\}$

$\mathcal{F}'$ is still a cover of E & each set in $\mathcal{F}'$ is open in E.

As E is compact in E, we can find $\mathcal{E} \subseteq \mathcal{F}$ (finite)

s.t. $\cup \mathcal{E}' \supseteq E$

say $\mathcal{E}' = \{U_1 \cap E, \ldots U_n \cap E\}$

Put $\mathcal{E} = \{U_1, \ldots U_n\}$ then $\mathcal{E} \subseteq \mathcal{F}$ (finite)

& $\cup \mathcal{E} \supseteq E$

so E is compact in X.

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① Thm:- (X,d) metric space, $E \subseteq X$

① Assume E is compact Then

ⓐ E is closed in X

ⓑ E is bounded in X

ⓒ for every infinite $A \subseteq E$.

$\exists\ x \in E$ s.t. x is a limit point of A in E.

$(A' \cap E \neq \phi)$

② If X is compact & E is closed in X, then E is compact.

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② Thm: Let $E \subseteq \mathbb{R}^n$, then

(i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) where

(i) E is closed in $\mathbb{R}^n$ & bounded

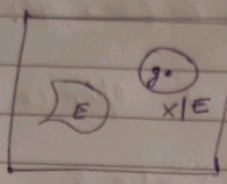
(ii) E is compact

(iii) for every infinite $A \subseteq E$. A has limit Point in E

Proof Thm (E) $\{B(x,n) : n \in \mathbb{N}\} \supseteq E$

E is compact in X but not closed in X.
Let $y \in E'$ s.t. $y \notin E$.

so $(\forall r>0)$ $(B(y,r) \cap E$ is infinite) — (1)



$\{X \setminus c(y, 1/n) : n \geq 1\}$

$\underline{\cup \{X \setminus c(y, 1/n) : n \geq 1\} \supseteq E}$

$\parallel$ (as $y \notin E$)

$X \setminus \{y\}$

E compact $\Rightarrow (\exists\, m \geq 1)\ (X \setminus c(y, 1/m) \supseteq E)$

$\Rightarrow B(y, \frac{1}{m}) \cap E = \phi$

that contradicts (1)

E is compact. Let $A \subseteq E$ be infinite. suppose ~~no~~ no point in E is in A. for each $x \in E$ fix $r>0$ s.t. Put $B(x, r_x) \cap A = \begin{cases} \{x\} & \text{if } x \in A \\ \phi & \text{otherwise} \end{cases}$

$\mathcal{F} = \{ B(x, r_x) : x \in E \}$ is an open cover of E

Let $\{ B(x_1, r_{x_1}), \ldots, B(x_n, r_{x_n}) \}$ be a finite subset of E

As $A \subseteq E$; $A \subseteq \underset{\text{finite}}{B(x_1, r_1) \cup \ldots} \underset{\text{finite}}{B(x_n, r_n)} \Rightarrow$

$\cap$ A $\quad\quad$ $\cap$ A

Impossible because A is infinite and $B(x_1,r_1) \cup \cdots \cup B(x_n,r_n)$ is finite.

Let $\mathcal{F}$ be an open cover of E (in a open sets in Y)

$\mathcal{F} = \mathcal{F} \cup \{X \setminus E\}$ is open cover of E

$\cup \mathcal{E}' \supseteq X$, $\mathcal{E} = \mathcal{E}' \setminus \{X \setminus E\}$ covers E

Thm II

Proof :→ (i) ⇒ (ii)

$E \subseteq \mathbb{R}^n$ closed and bounded

Bounded ⇒ $(\exists K>0)\ (E \subseteq [-K,K] \times \cdots \times [-K,K])$

E closed; $[-K,K] \times \cdots \times [-K,K]$ compact

By using Thm I ②

(ii) ⇒ (iii) By Thm I ① ©

(iii) ⇒ (i) Assume (iii) E is bounded it's easy.

If not, choose

$A = \{x_k : k \geq 1\} \subseteq E$

$\|x_k\| \geq k$

A has no limit pt. in E

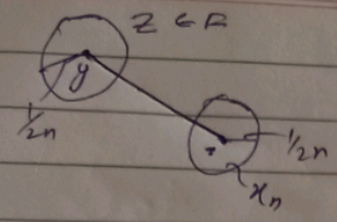
E is closed :- suppose not

Fix $y \in E' \setminus E$ choose

$\langle x_n : n \geq 1 \rangle$ s.t. $d(y, x_n) < \frac{1}{n}$

Put $A = \{x_n : n \geq 1\}$

H.W. $[a_1, b_1] \times \cdots [a_n, b_n] \subseteq \mathbb{R}^n$ is compact.



$y \in E'$
$y \notin E$

why A has not limit pt. in A?

say $z \in$